



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 06

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Program:

```
#include <stdio.h>

void merge(int a[], int l, int m, int r)
{
    int i = l, j = m + 1, k = 0;
    int temp[100];
    while (i <= m && j <= r)
    {
        if (a[i] < a[j])
            temp[k++] = a[i++];
        else
            temp[k++] = a[j++];
    }
    while (i <= m)
        temp[k++] = a[i++];
    while (j <= r)
        temp[k++] = a[j++];
    for (i = l, k = 0; i <= r; i++, k++)
        a[i] = temp[k];
}
```

```
void mergeSort(int a[], int l, int r)
{
    if (l < r)
    {
        int m = (l + r) / 2;
        mergeSort(a, l, m);
        mergeSort(a, m + 1, r);
        merge(a, l, m, r);
    }
}

int main()
{
    int a[] = {38, 27, 43, 3, 9, 82, 10};
    int n = 7, i;
    mergeSort(a, 0, n - 1);
    printf("[");
    for (i = 0; i < n; i++)
    {
        printf("%d", a[i]);
        if (i < n - 1)
            printf(", ");
    }
    printf("]");

    return 0;
}
```

Output:

```
mainc Run ▶ Output
3 void merge(int a[], int l, int m, int r)
4 {
5     int i = l, j = m + 1, k = 0;
6     int temp[100];
7
8     while (i <= m && j <= r)
9     {
10         if (a[i] < a[j])
11             temp[k++] = a[i++];
12         else
13             temp[k++] = a[j++];
14     }
15
16     while (i <= m)
17         temp[k++] = a[i++];
18
19     while (j <= r)
20         temp[k++] = a[j++];
21
22     for (i = l, k = 0; i <= r; i++, k++)
23         a[i] = temp[k];
24 }
25
26 void mergeSort(int a[], int l, int r)
27 {
28     if (l < r)
29     {
30         int m = (l + r) / 2;
31         mergeSort(a, l, m);
32         mergeSort(a, m + 1, r);
33         merge(a, l, m, r);
34     }
35 }
36
37 int main()
38 {
39     int a[] = {38, 27, 43, 3, 9, 82, 10};
40     int n = 7, i;
41 }
```

[[19,22],[43,50]]